Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

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Submitted By:- G13

***Abstract*—Automatic visual inspection of woven fabric requires finding small local defects on a repetitive background. We present a classical, training-free pipeline for the TILDA textile database. Fabric images are examined with point processing, spatial filtering and 2D Fourier analysis. Each image is then split into 32×32 windows described by gray-level co-occurrence matrix (GLCM) statistics and uniform local binary patterns (LBP). Each window receives a z-score anomaly score relative to a defect-free reference, the GLCM and LBP scores are fused, and an adaptive threshold $\mu + k\sigma$ marks anomalous windows, which are merged into defect regions by connected-component analysis. On textile 1 the line-like e4 defect produces a contiguous band of anomalous windows (62 windows, 16.1%) against 16 (4.2%) for the normal image. A single reference fails across textiles (100% false alarms on classes 2–4); class-specific references bring normal-image alarms to 2.3–7.0% and raise the anomalous fraction on e4 images of all four classes (7.0–18.5%), although clear coarse localization is obtained only for the tested textile-1 sample.**

***Index Terms*—fabric defect detection, texture analysis, GLCM, local binary patterns, Fourier transform, anomaly localization**

I. Introduction

Defect detection in textiles is a standard industrial inspection task. Deep networks need many labeled defects, which are scarce, while the defect-free texture is easy to collect. Prior work has used LBP [1], GLCM [2] and their combination [3] for fabric defects, and has studied decision fusion [4] and GLCM–Gabor comparisons [5]. A fixed decision threshold transfers poorly between fabrics with different texture and brightness, so we make the threshold adaptive. This motivates a one-class approach: model the normal texture statistically and flag windows that deviate from it. This project applies the classical Digital Image Processing toolbox covered in ECE 501 (intensity transformations, spatial filtering, the 2D DFT and frequency-domain filtering [6]) together with two texture descriptors, GLCM [7] and LBP [8], to the TILDA database [9]. The contributions are: (i) a comparison of spatial- and frequency-domain views of normal and defective fabric; (ii) a fused GLCM+LBP window-level anomaly score with an adaptive threshold; (iii) pixel-level localization via connected components; and (iv) a validation across textile classes. All code is Python with NumPy, OpenCV and scikit-image.

II. Methodology

A. Dataset

We use the TILDA textile database [9], subset CD1+CD2: 3270 TIFF files, of which 3148 follow the naming convention and are used. File names follow the pattern `cXrXeXnX.tif`, giving textile class, representative, error class (e0 = defect-free) and image number; metadata are parsed with a regular expression. Images are 8-bit gray-scale, 512×768 pixels. There are four textile classes and eight error classes (e0 = defect-free, e1–e7 = defect types); each class–error combination has 100 images, except three (c2e2: 99, c4e2: 50, c4e5: 99), giving 400 normal images. The development pair is textile 1, representative 1: a normal image (e0) against an e4 image. The validation set has one e0 and one e4 image from each of the four textile classes (eight images).

B. Pre-analysis in the spatial and frequency domains

We applied gamma correction ($\gamma \in \{0.5, 1, 1.5\}$), min–max contrast stretching, histogram equalization, Gaussian smoothing (3×3, 5×5), the Laplacian, and Sobel gradients with magnitude $\sqrt{G_x^2 + G_y^2}$. In the frequency domain, $F(u,v) = \sum_{x,y} f(x,y) e^{-j2\pi(ux/M + vy/N)}$ was computed with `fft2` and `fftshift`, and displayed as $\log(1 + |F|)$ together with the phase. Ideal low- and high-pass masks with $D \le D_0$ and $D > D_0$, $D_0 = 30$, were applied as $G = HF$ and inverted.

C. Window features

Each image is divided into non-overlapping 32×32 windows (a 16×24 grid, 384 windows).

GLCM. Each window is quantized to 32 gray levels; a symmetric, normalized GLCM is built for distance 1 and angles $0°, 45°, 90°, 135°$. Contrast, homogeneity, energy and correlation are averaged over the angles [7].

LBP. Uniform LBP with $P = 8$, $R = 1$ gives a 10-bin normalized histogram per window [8].

Image → 32 × 32 windows → GLCM / LBP features → z-score vs. normal reference → fused score S → threshold $\mu + k\sigma$ → mask → connected components / boxes

Fig. 1. Overview of the detection pipeline.

D. Anomaly score and threshold

From the normal reference image we compute the mean μ_i and standard deviation σ_i of each feature. A window with feature vector $\mathbf{x}$ gets

$$S_G = \sqrt{\sum_{i=1}^{4} \left(\frac{x_i - \mu_i}{\sigma_i}\right)^2}, \quad S_L = \sqrt{\sum_{i=1}^{10} \left(\frac{h_i - \mu_i}{\sigma_i}\right)^2}, \quad (1)$$

for the GLCM and LBP descriptors. Both are standardized with the normal-image score statistics and fused:

$$S = \sqrt{z_G^2 + z_L^2}, \qquad T = \mu_S + k\,\sigma_S, \quad (2)$$

where μ_S, σ_S are the mean and standard deviation of S over normal windows and $k \in \{1.5, 2.0, 2.5\}$. We select $k = 2$. For validation, scores are standardized per textile class with fixed normal-image statistics, so the threshold becomes $T = \sqrt{2}\,k \approx 2.83$.

E. Localization

Windows with $S > T$ are expanded to a pixel mask (each window to 32×32 pixels). Eight-connected components [6] give area, centroid and bounding box of each candidate defect.

III. Results

A. Spatial and frequency analysis

Fig. 2 shows that e4 is a thin dark line crossing the fabric, whereas other classes are blotches, shading changes or texture changes. Global intensity statistics (textile 1, 100 images per error class) barely separate the classes: the mean ranges from 123.8 (e6) to 158.6 (e7), e4 has mean 134.8 and standard deviation 14.3 against 127.8 and 12.6 for e0, and classes such as e1, e2 and e7 deviate more in mean than e4 does. Because global brightness varies between images, a local texture analysis is needed.

Sobel gradients (Step 19) respond strongly along both edges of the e4 line; the gradient-magnitude standard deviation rises from 15.3 (normal) to 19.0 (e4) and its maximum from 147 to 233, while the mean stays at 28.4 vs. 28.6.

In the Fourier domain (Fig. 3) the normal fabric shows discrete, regularly placed peaks, the signature of the periodic weave. The e4 spectrum keeps these peaks but adds an oriented streak through the DC term and a brighter horizontal axis, typical of a straight edge. The ideal low-pass result keeps the line as a blurred band with visible ringing, while the high-pass result shows the line as a bright band on a dark background. Frequency filtering therefore shows the defect but does not by itself indicate where it lies in a quantitative way, which motivates the window-level analysis below.

TABLE I
Normal GLCM reference (textile 1, e0, 384 windows)

Feature	Mean	Std
Contrast	0.5333	0.0780
Homogeneity	0.7675	0.0221
Energy	0.4052	0.0347
Correlation	0.5654	0.0639

TABLE II
Anomaly score statistics, normal vs. defect e4 (textile 1)

	Normal		Defect e4	
Score	Mean	Std	Mean	Std
GLCM	1.752	0.961	2.875	2.807
LBP	3.058	0.790	3.942	2.170
Fused S	1.215	0.722	2.404	3.593

B. Window-level anomaly scores

Table I lists the normal GLCM reference and Table II the score statistics. Defect windows have a higher mean and a much larger spread than normal windows (Fig. 4): the GLCM score reaches 16.3 and the LBP score 15.5 in the defect image, against about 5 or less in the normal image. The mean LBP histograms of the two images are close (largest bin difference 0.010, Step 45), so the evidence lies in a few windows rather than in the average pattern.

C. Adaptive threshold and localization

The fused score on the normal image has $\mu_S = 1.2146$ and $\sigma_S = 0.7217$, so $k = 2$ gives $T = 2.658$. Table III shows the trade-off: a larger k lowers normal-image alarms but also defect detections. We use $k = 2$.

The heatmap (Fig. 5) shows a diagonal band of high scores along the e4 line (maximum 20.4), while the normal image has scattered values up to 6.0, mostly near its borders. Fig. 6 shows the masks and connected components. The defect mask contains 62 windows (63 488 pixels) in seven components: one large component of 55 296 pixels (54 windows) following the line, one of 3 072 pixels and five isolated windows. The normal mask contains 16 windows in seven components, five of them isolated single windows.

D. Validation across textile classes

Table IV compares two scoring setups on one normal and one e4 image per textile class. With the textile-1 reference (Step 63) the normal images of classes 2, 3 and 4 are flagged in 100% of windows (mean fused score 54.9, 23.9 and 67.6), so one reference cannot serve all fabrics. With class-specific references (Steps 64–68) normal-image alarms fall to 2.3–7.0% and the e4 images always have a larger mean score and a larger anomalous fraction than the normal ones. Because each class-specific reference is constructed from the same normal image used for evaluation, the reported normal alarm rates should be regarded as optimistic; held-out normal images are required for unbiased estimation.

The anomaly maps (Fig. 7) show a clear band along the defect for textile 1 only. For classes 2–4 the defect images

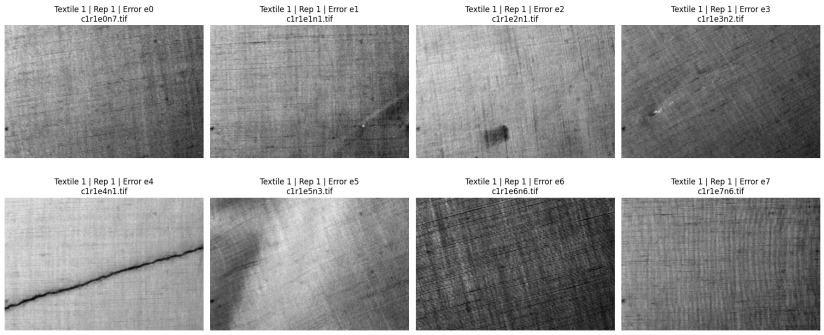


Fig. 2. Textile 1, representative 1: one image per error class, e0 (defect-free) to e7.

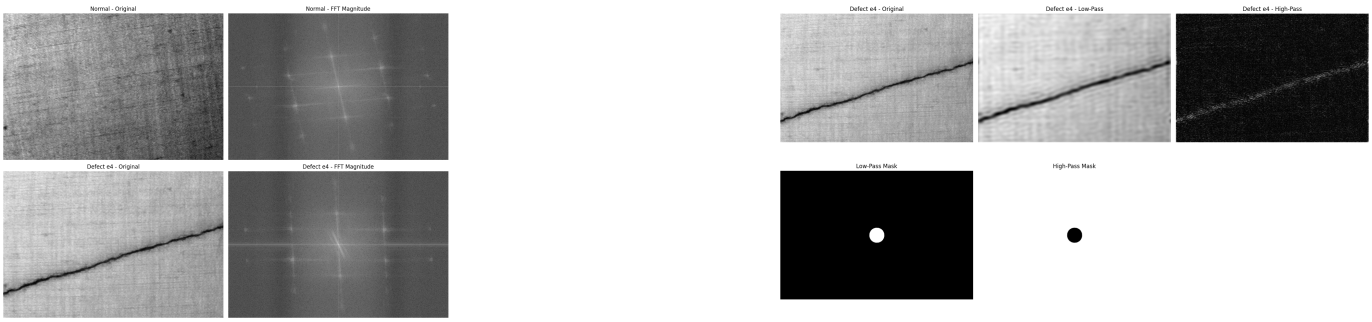


Fig. 3. Left: normal and e4 images with log-magnitude spectra. Right: e4 image, ideal low-pass and high-pass results ($D_0 = 30$) and the masks.

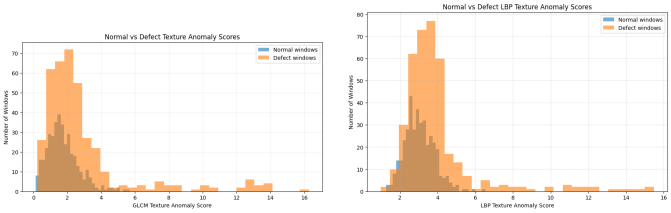


Fig. 4. Window-score histograms, normal vs. defect: GLCM (left), LBP (right).

TABLE III
EFFECT OF k ON ANOMALOUS WINDOWS (TEXTILE 1, 384 WINDOWS)

k	T	Normal	Defect e4
1.5	2.297	23 (6.0%)	77 (20.1%)
2.0	2.658	16 (4.2%)	62 (16.1%)
2.5	3.019	11 (2.9%)	53 (13.8%)

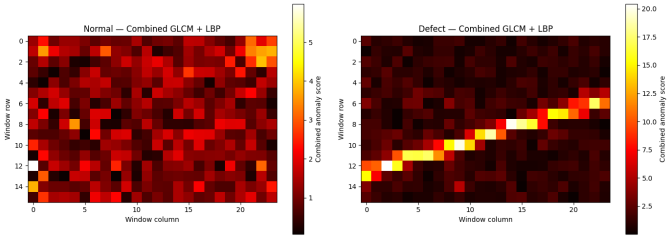


Fig. 5. Fused anomaly score per window (16×24 grid), normal (left) and e4 (right).

contain a few brighter windows but also many comparable ones elsewhere, and the maps are not clearly more structured than the normal maps.

TABLE IV
VALIDATION (N = NORMAL, D = E4 DEFECT), $T = \sqrt{2}\,k$ FOR CLASS-SPECIFIC SCORING

Class	Fixed ref. anom.% N/D	Class ref. mean S N/D	Class ref. anom.% N/D
1	3.9 / 16.1	1.23 / 2.68	2.3 / 14.6
2	100 / 100	1.22 / 1.40	4.2 / 7.0
3	100 / 100	1.18 / 1.52	4.9 / 9.9
4	100 / 99.5	1.12 / 1.70	7.0 / 18.5

IV. DISCUSSION

For textile 1 the method behaves as intended: the e4 line gives high scores in a connected set of windows, which merge into one dominant region. The score histograms (Fig. 4) show that the defect distribution exhibits a pronounced upper tail, not a shift of the whole distribution, so a threshold based on the normal mean and standard deviation is appropriate. Fusion of the descriptors helps because the mean LBP histograms of the two images almost coincide.

The textile-class validation shows the main weakness. A fixed reference gives 100% false alarms on classes 2–4, so references must be class-specific. Even then, only class 1 gives clear localization. The e4 defect in class 2 raises the anomalous fraction only from 4.2% to 7.0%, and the other classes (periodic gingham in class 3, strongly patterned print in class 4) have high texture variation between windows, so the defect does not stand out against it. The Fourier analysis points to a possible remedy: periodic fabrics have discrete spectral peaks, so removing these peaks (notch filtering) before feature extraction could suppress the regular pattern and leave the defect.

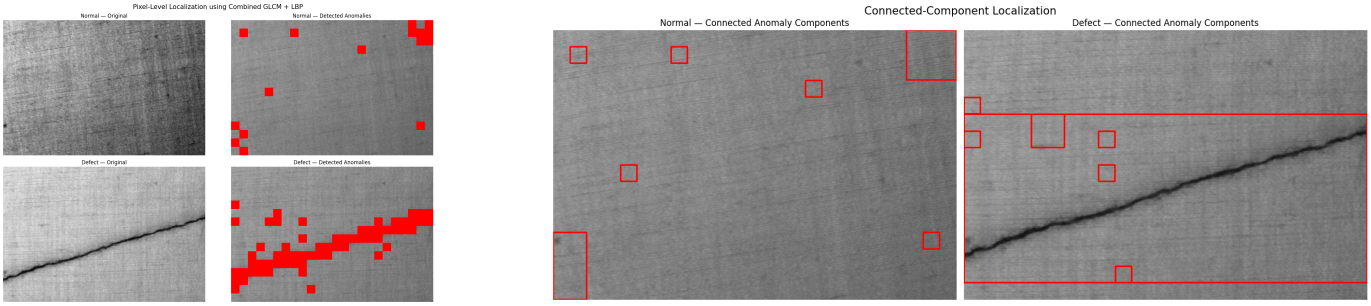


Fig. 6. Left: detected anomalous windows (red) for the normal and e4 images. Right: bounding boxes of connected components.

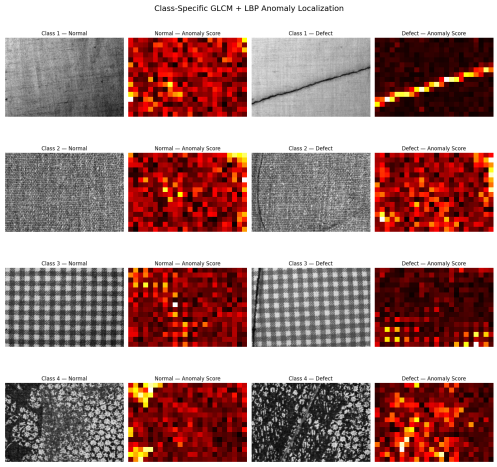


Fig. 7. Class-specific anomaly maps for the normal and e4 images of the four textile classes.

Limitations. (i) Each reference is built from one normal image and that image is also evaluated against it, so normal false-alarm rates are optimistic; held-out normal images are needed. (ii) Validation uses a single e4 image per class and a single error class, so the results are indicative, not statistical. (iii) A 32×32 window limits localization to window granularity, and a bounding box around a diagonal defect is loose: the dominant defect component in Fig. 6 spans the full image width (768×320 px) although the defect is a thin line. (iv) A single k is used and no ROC or precision/recall against ground-truth masks is reported. Future work: overlapping windows, held-out normals, all error classes e1–e7, frequency-domain preprocessing and ROC/F1 analysis.

V. Conclusion

A training-free GLCM+LBP pipeline with an adaptive $\mu + k\sigma$ threshold detects and localizes a line-like fabric defect on TILDA textile 1 at window level, using only defect-free reference statistics: 62 of 384 windows (16.1%) are flagged in the defect image versus 16 (4.2%) in the normal one, and connected-component analysis returns the defect as one dominant region. A preliminary validation across the four textile classes shows that class-specific references are necessary (a single reference flags every window of three classes), and the e4 images always show a higher anomalous fraction (7.0–18.5% against 2.3–7.0%); however, reliable localization for strongly periodic or patterned fabrics would need further work, such as frequency-domain suppression of the pattern.

References

[1] F. Tajeripour, E. Kabir, and A. Sheikhi, "Fabric defect detection using modified local binary patterns," *EURASIP J. Adv. Signal Process.*, 2007.
[2] X. Zhang and X. Fan, "Fabric defect detection based on GLCM approach," ICIMMS, 2016.
[3] L. Zhang, J. Jing, and H. Zhang, "Fabric defect classification based on LBP and GLCM," *J. Fiber Bioeng. Inform.*, vol. 8, no. 1, pp. 81–89, 2015.
[4] A. S. Tolba *et al.*, "Decision fusion for visual inspection of textiles," *Textile Res. J.*, 2010.
[5] "Fabric defect detection based on GLCM and Gabor filter: A comparison," *Optik*, 2013.
[6] R. C. Gonzalez and R. E. Woods, *Digital Image Processing*, 4th ed. Pearson, 2018.
[7] R. M. Haralick, K. Shanmugam, and I. Dinstein, "Textural features for image classification," *IEEE Trans. Syst., Man, Cybern.*, vol. SMC-3, no. 6, pp. 610–621, 1973.
[8] T. Ojala, M. Pietikäinen, and T. Mäenpää, "Multiresolution gray-scale and rotation invariant texture classification with local binary patterns," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 24, no. 7, pp. 971–987, 2002.
[9] TILDA Textile Texture Database, Workgroup on Texture Analysis of the DFG, 1996.